

Priyanka Chatterjee

SDE III · Workflow AI · Custom AI Agents & RAG · Node.js/JavaScript · 9,300 ★

OSS Contributor · 3.5+ Years

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SUMMARY

Software Development Engineer III - Workflow **AI** with 3.5+ years of experience and expertise in **JavaScript/Node.js** backend development, **AI-powered automation**, **custom AI agents**, and Retrieval-Augmented Generation (**RAG**) systems. Built a **RAG-based AI** agent with a vector-database-backed **knowledge base** and a multi-agent **automation** workflow at Turbot HQ, cutting resource-discovery latency 70-90% through **AI-driven automation**.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Python

AI Engineering: LLM Application Development, Agentic Workflows, Multi-Agent Systems, Custom AI Agents, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), OpenAI Agents SDK, Pydantic AI, Vector Storage (AWS S3), Prompt Engineering

AI Tooling & Multi-Model Integration: Claude Code, Cursor, Logfire (LLM observability), multi-model switching (OpenAI, Gemini, Anthropic, Mistral)

Backend & Cloud: Node.js, Serverless (AWS Lambda), Azure Event Grid, API Integration (OpenAI, Slack Bot API)

Databases & Data: DuckDB, Parquet, MongoDB, Vector Storage (AWS S3)

WORK EXPERIENCE

Turbot HQ, Inc. Remote

August 2022 – April 2026

Software Engineer I

- Designed a multi-agent workflow on the **OpenAI** Agents SDK to orchestrate **custom AI agents** that generate cloud preventive-policy **automation** commands, validate them in a live test environment, self-correct failures, and score outcomes against an internal metric — an end-to-end **AI-powered automation** pipeline for complex tasks.
- Built a **RAG-based** internal Pydantic **AI** agent with a **knowledge base** spanning documentation and code for all internal repositories, ingested into S3 vector storage; deployed to AWS Lambda and integrated with the Slack Bot API to deliver context-aware, **AI-driven** answers to cross-team queries.
- Authored an LLM-driven Claude Code skill automating code refactoring, unit test generation, deployment, and PR creation — integrating **AI** directly into code-execution **automation** actions; used it to migrate 35+ Azure cloud-governance modules to event-driven eventing, cutting resource-discovery latency 70-90%.
- Led the AWS SDK for **JavaScript** v2 to v3 migration across 180+ call sites and 11+ AWS services in production **Node.js** Lambdas; reduced policy-code duplication ~60% via templating, keeping the platform supported past AWS SDK v2 EOL.
- Engineered core features for Tailpipe, a CLI on DuckDB + Parquet, lifting log-processing throughput 30% to 1,000+ logs per second by chunking and batching file reads and streaming writes at **scale**; reduced memory use to eliminate out-of-memory failures.
- Mentored teammates on adopting **AI-assisted** engineering tools (Claude Code, Cursor) into their daily **workflows**.

PROJECTS

General-Purpose AI Chatbot / Agent (Model Context Protocol) (Jun 2025 – Aug 2025)

Tools framework enabling the creation of **custom AI agents**, built with Pydantic **AI**: configurable external MCP servers via a persistent MongoDB-backed registry, JSON config loader with strict input validation, per-server failure isolation, native tool calling, mid-conversation multi-model switching (**OpenAI**, Gemini, Anthropic, Mistral), conversation-history persistence, sync + streaming output, prompt-injection safeguards, and Logfire observability.

EDUCATION

University of Engineering & Management (UEM) Kolkata, WB, IN

August 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.28

Relevant Coursework: Object-Oriented Programming, DBMS, Data Structures and Algorithms, Shell Scripting in Linux, Low-Level Design, High-Level Design